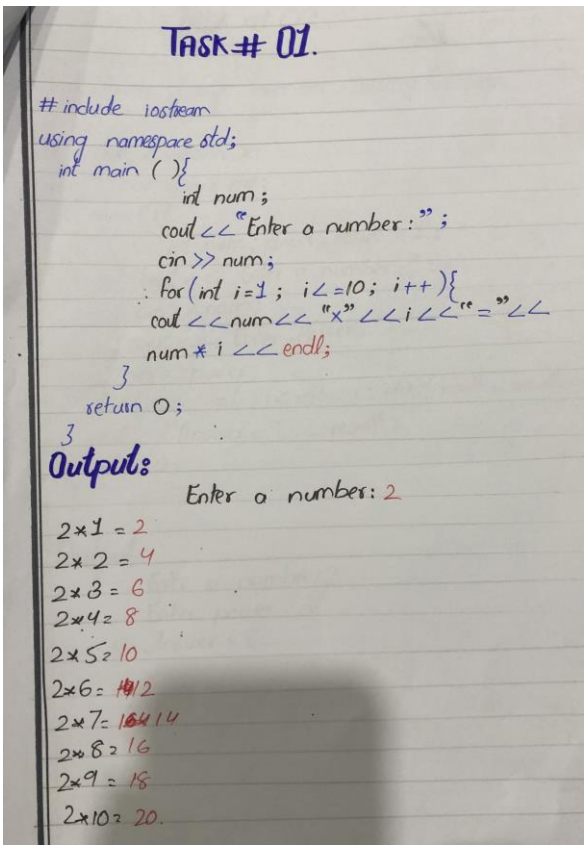
Programming Fundamentals

Lab 5 Submission	
Name	| Saleha Amjad |
Roll No	| TO BE ADDED |
Program	| BSCS – 1st Semester |
Section	(E)
Submitted To	|Sir Asad Majeed|
Date	| 10 – 10 – 26 |

Lab Task 01

Taskl: C++ program to Print Table of any Number.

HANDWRITTEN :



CODE On Desktop :

Task 1.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    int num;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= 10; i++) {
        cout << num << " x " << i << " = " << num * i << endl;
    }

    return 0;
}

```

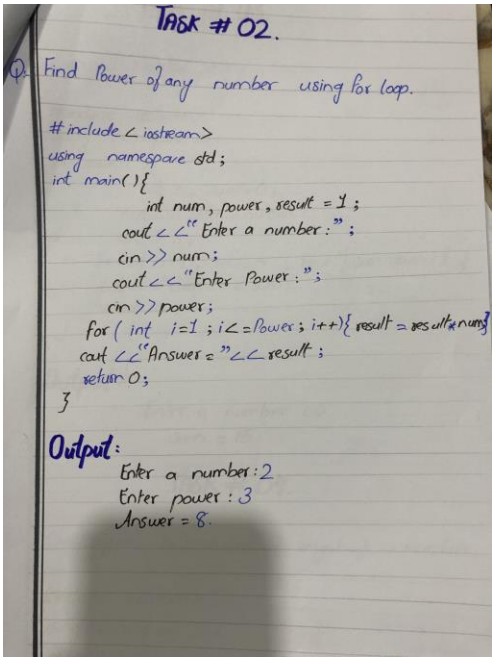
OUTPUT :

```
Enter a number: 2
2 x 1 = 2
2 x 2 = 4
2 x 3 = 6
2 x 4 = 8
2 x 5 = 10
2 x 6 = 12
2 x 7 = 14
2 x 8 = 16
2 x 9 = 18
2 x 10 = 20
```

Lab Task 02

Find power Of any number using for loop.

HANDWRITTEN :



TASK # 02.

Q. Find Power of any number using for loop.

```
#include<iostream>
using namespace std;
int main(){
        int num, power, result = 1;
        cout<<"Enter a number:";
        cin>>num;
        cout<<"Enter Power:";
        cin>>power;
    for( int i=1 ; i<=Power; i++){ result = result*num;
    cout<<"Answer = "<<result;
    return 0;
}
```

Output:

Enter a number: 2
Enter power: 3
Answer = 8.

CODE On Desktop :

Task 2.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    int num, power, result = 1;

    cout << "Enter a number: ";
    cin >> num;

    cout << "Enter power: ";
    cin >> power;

    for (int i = 1; i <= power; i++) {
        result = result * num;
    }

    cout << "Answer = " << result;

    return 0;
}

```

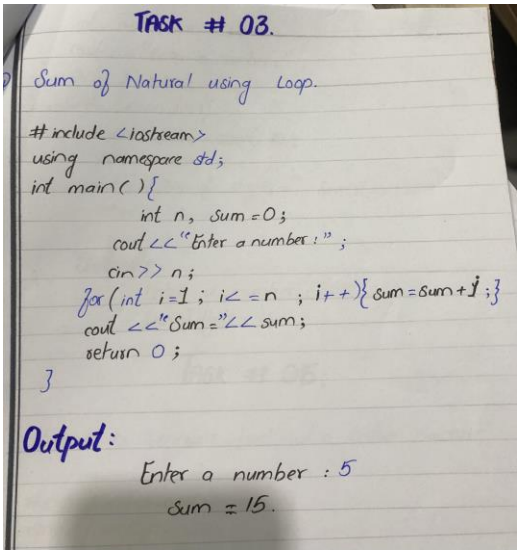
OUTPUT :

```
Enter a number: 2
Enter power: 3
Answer = 8
```

Lab Task 03

Sum of Natural Numbers using loop.

HANDWRITTEN :



CODE On Desktop :

Task 3.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;

    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        sum = sum + i;
    }

    cout << "Sum = " << sum;

    return 0;
}

```

OUTPUT :

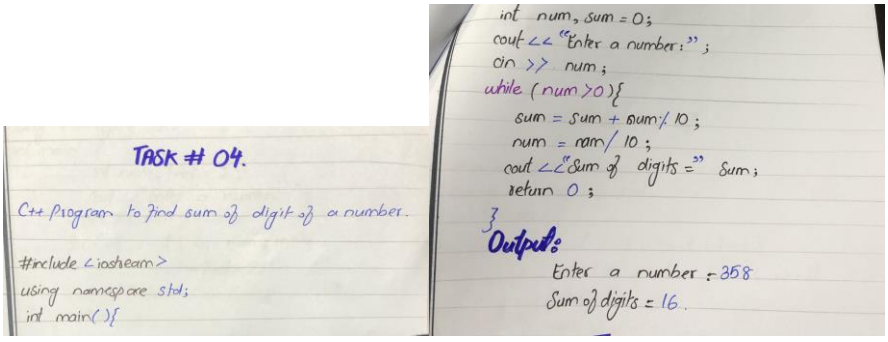
```
Enter a number: 5
Sum = 15
```

Lab Task 04

C++ program to find Sum Of digits Of a number.

Sum Of digits means add all the digits Of any number, for example we take any number like 358. Its sum Of all digit is 3+5+8=16. Using given code we can easily write c++ program.

HANDWRITTEN :



CODE On Desktop :

```cpp

#include <iostream>
using namespace std;

int main() {
    int num, sum = 0;

    cout << "Enter a number: ";
    cin >> num;

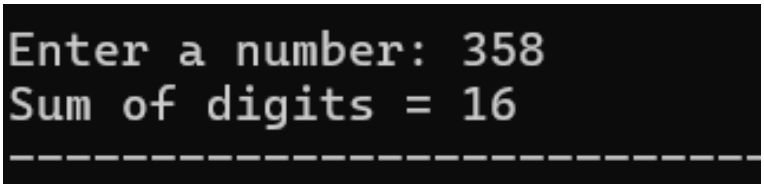
    while (num > 0) {
        sum = sum + num % 10;
        num = num / 10;
    }

    cout << "Sum of digits = " << sum;

    return 0;
}

```

OUTPUT :

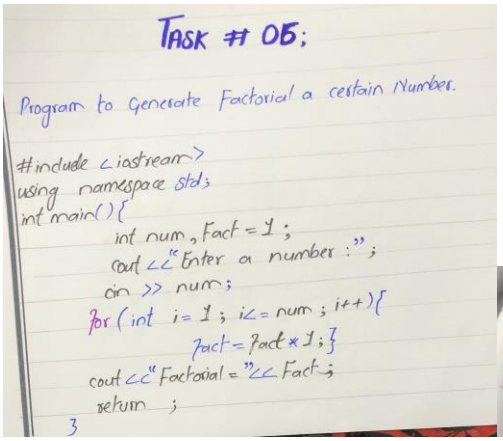


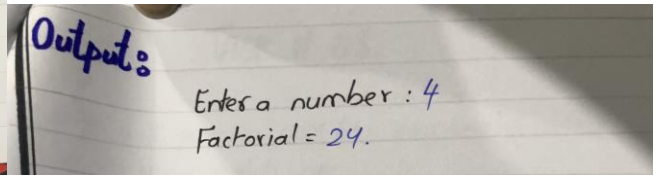
Lab Task 05

Program to Generate Factorial a Certain Number

Factorial of any number is the product of an integer and all the integers below it for example factorialof4 is 4!=4 * 3 * 2 * 1 = 24

HANDWRITTEN :





CODE On Desktop :

Task 5.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    int num;
    int fact = 1;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= num; i++) {
        fact = fact * i;
    }

    cout << "Factorial = " << fact;

    return 0;
}

```

OUTPUT :

```
Enter a number: 4
Factorial = 24
```

Lab Task 07

Write a C++ program that will print the following pattern.

*

**

***

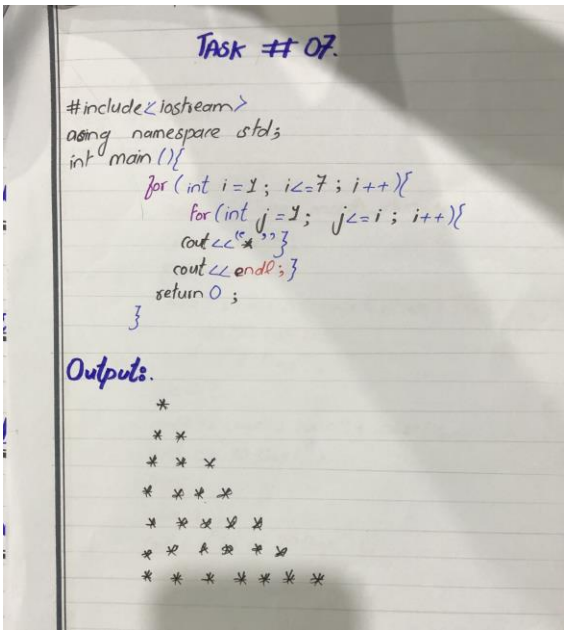
****

*****

******

****

HANDWRITTEN :



CODE On Desktop :

Task 6.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    for (int i = 1; i <= 7; i++) {
        for (int j = 1; j <= i; j++) {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}

```

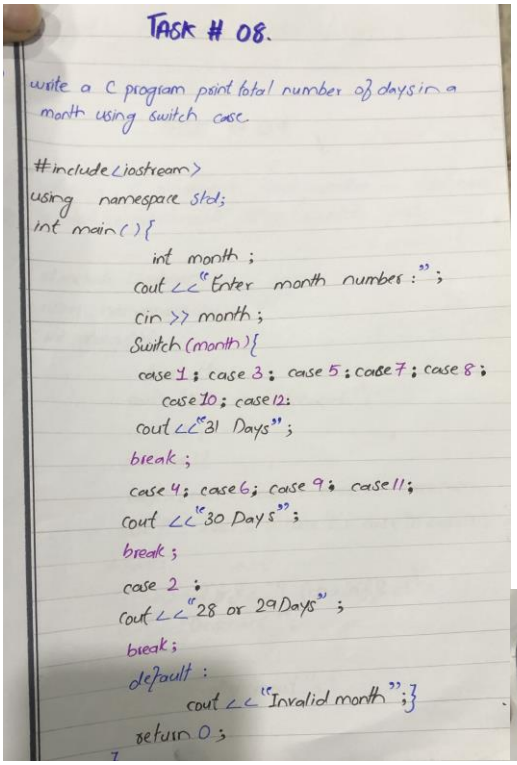
OUTPUT :

```
*
**
***
****
*****
******
*******
```

Lab Task 08

Write a C program print total number Of days in a month using switch case.

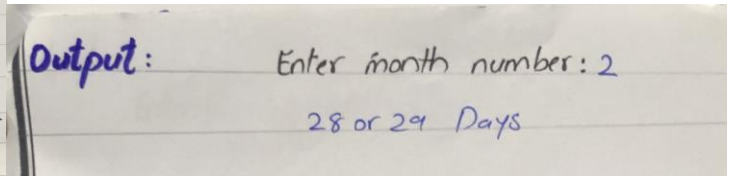
HANDWRITTEN :



TASK # 08.

write a C program print total number of days in a month using switch case.

```
#include<iostream>
using namespace std;
int main(){
        int month ;
        cout<<"Enter month number :";
        cin >> month ;
        Switch (month){
        case 1 ; case 3 ; case 5 ; case 7 ; case 8 ;
          case 10 ; case 12:
        cout<<"31 Days";
        break ;
        case 4 ; case 6 ; case 9 ; case 11 ;
        cout <<"30 Days";
        break ;
        case 2 ;
        cout<<"28 or 29 Days" ;
        break ;
        default :
              cout<<"Invalid month";}
        return 0 ;
```



Output : Enter month number : 2

28 or 29 Days

CODE On Desktop :

Task 7.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    int month;

    cout << "Enter month number (1-12): ";
    cin >> month;

    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << "31 days";
            break;

        case 4: case 6: case 9: case 11:
            cout << "30 days";
            break;

        case 2:
            cout << "28 or 29 days";
            break;

        default:
            cout << "Invalid month";
    }

    return 0;
}

```

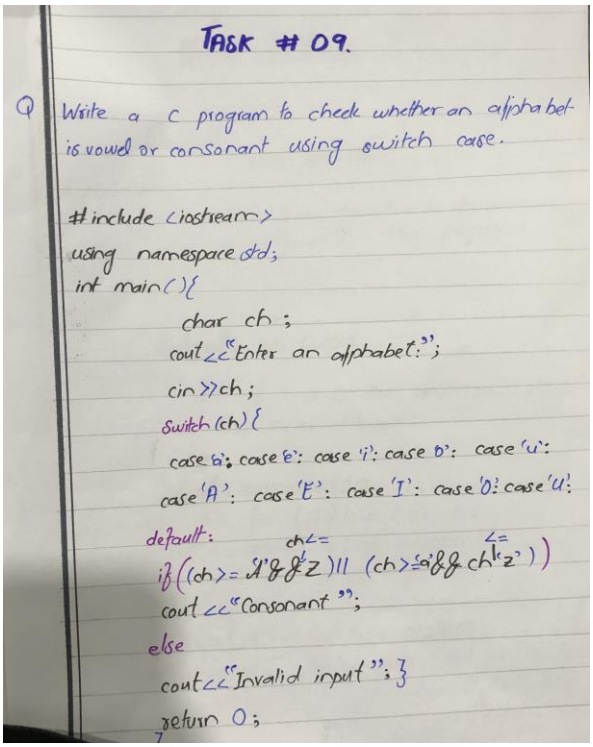
OUTPUT :

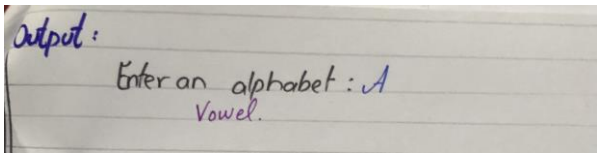
```
Enter month number (1-12): 2
28 or 29 days
```

Lab Task 09

Write a C program to check whether an alphabet is vowel or consonant using switch case.

HANDWRITTEN :





CODE On Desktop :

Task 8.cpp

```cpp

#include <iostream>
using namespace std;

int main() {
    char ch;

    cout << "Enter an alphabet: ";
    cin >> ch;

    switch (ch) {
        case 'a': case 'e': case 'i':
        case 'o': case 'u':
        case 'A': case 'E': case 'I':
        case 'O': case 'U':
            cout << "Vowel";
            break;

        default:
            if ((ch >= 'a' && ch <= 'z') ||
                (ch >= 'A' && ch <= 'Z'))
                cout << "Consonant";
            else
                cout << "Invalid input";
    }

    return 0;
}

```

OUTPUT :

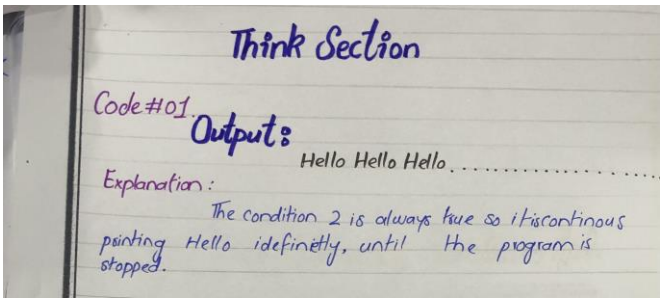
```
Enter an alphabet: A
Vowel
```

THINK SECTION ! | Program 01

What will be the output of the C program?

```
#include<stdio.h>
int main()
{
        for(5;2;2)
        {
                cout<<"Hello";
        }
        return 0;
}
```

HANDWRITTEN :



CODE On Desktop :

think1.cpp

```
1  #include<iostream>
2  using namespace std;
3  int main(){
4      for(5;2;2){
5          cout<<"Hello";
6      }
7      return 0;
8  }
```

OUTPUT :

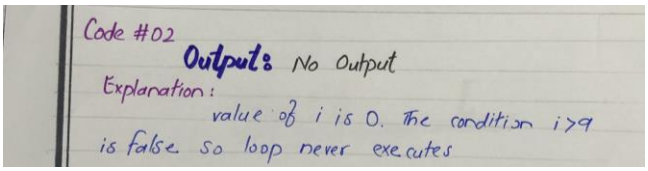
```
HelloHelloHelloHelloHelloHelloHell
HelloHelloHelloHelloHelloHelloHell
HelloHelloHelloHelloHelloHelloHell
HelloHelloHelloHelloHelloHelloHell
HelloHelloHelloHelloHelloHelloHell
HelloHelloHelloHelloHelloHelloHell
Hello
```

Program 02

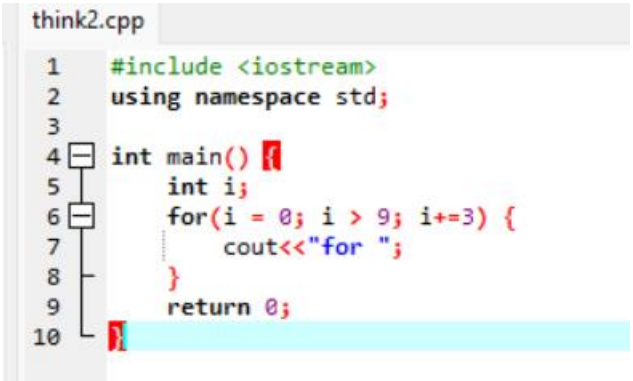
What will bethe output of the C program, if input is 6?

```
#include<stdio.h>
int main()
{
        int i;
        for(i = 0; i>9; i+=3)
        {
                cout<<"for ";
        }
        return 0;
}
```

HANDWRITTEN :



CODE On Desktop :



OUTPUT :

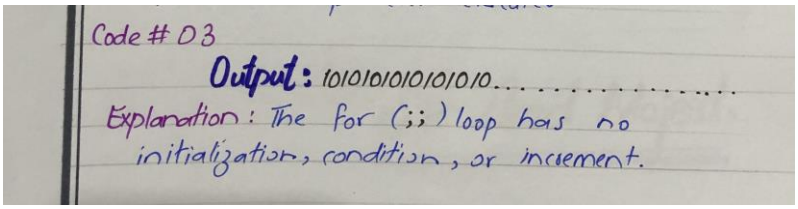
```
_______________________________
```

Program 03

What will bethe output of the C++ program.

```
#include<stdio.h>
int main()
{
        for(;;)
        {
                cout<<”10”;
        }
        return 0;
}
```

HANDWRITTEN :



CODE On Desktop :

think3.cpp

```
1   #include <iostream>
2   using namespace std;
3
4   int main() {
5       for(;;) {
6           cout<<"10";
7       }
8       return 0;
9   }
```

OUTPUT :

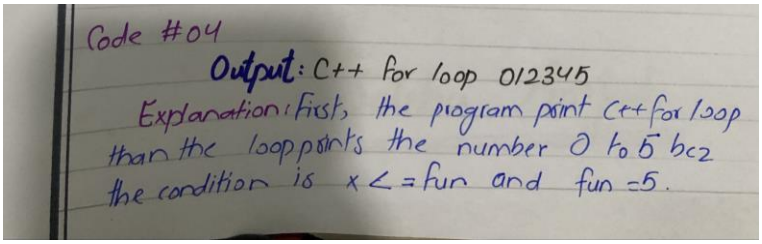
```
10101010101010101010101010101
10101010101010101010101010101
10101010101010101010101010101
```

Program 04

What will be the output of the C program?

```
int main()
{
        int fun=5;
        cout<<"C++ for loop ";
        int x = 5;
        for(x=0;x<=fun;x++)
```

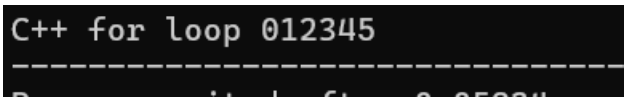
HANDWRITTEN :



CODE On Desktop :

```
1   #include <iostream>
2   using namespace std;
3
4   int main() {
5       int fun=5;
6       cout<<"C++ for loop ";
7       int x = 5;
8       for(x=0; x<=fun; x++) {
9           cout<<x;
10      }
11      return 0;
12  }
```

OUTPUT :



FIAmanAllah !